Z-Transform (applicable for discrete time intervals only)

Time ⟷ Freq

Why to LT

CTFT ⟷ LT (generalization of CTFT)

$j\omega \longleftrightarrow s = \sigma + j\omega$

(enlarges the capability of signals)

Why

DTFT ⟷ Z-Transform (generalization of DTFT)

$$X(\omega) = \sum_{n=-\infty}^{+\infty} x(n) e^{-j\omega n}$$

→ $x(n)$ must be summable to Exist $X(\omega)$

$$e^{j\omega} \longleftrightarrow re^{j\omega} = z$$

If $r = 1$, we get DTFT,
If $r \neq 1$ we get other Transform

→ In LT we worked with rectangular Coordinates
ie $\sigma + j\omega$

⇒ where as in Z.T, we work with polar coordinates
'r' distance from origin
'ω' +ve angle wrt to +ve real axis

→ $$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

provided $x(n)$ is summable

*

→ Depending upon value of 'r', it may converge or it may not

→ Z-Transform is exactly equal to the FT, at $z = e^{j\omega}$

Instead of 'ω' we replace by $e^{j\omega}$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Given $x(n)$ then

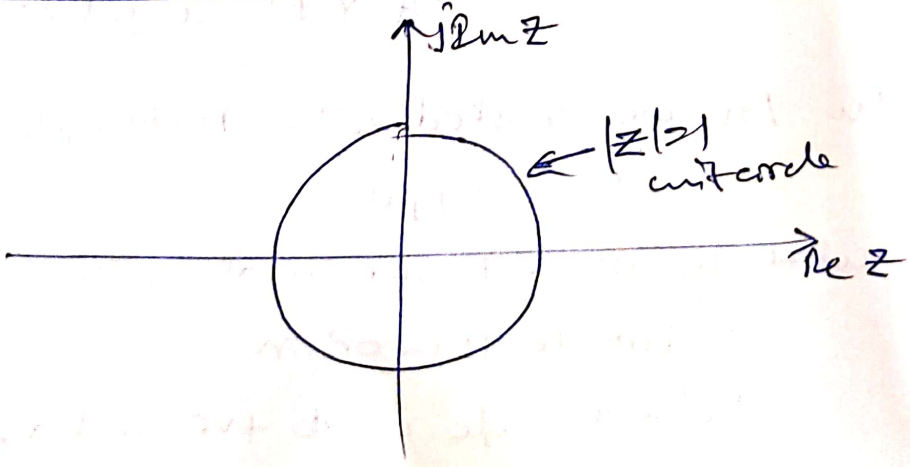
$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = Z[x(n)]$$

$$x(n) \xleftrightarrow{Z} X(z)$$

→ If FT Exist, then it must $r = 1$

$$\boxed{F[x(n)] = Z[x(n)]_{z=e^{j\omega}}}$$

Style of representing Z-plane :- $z = re^{j\theta}$, in general



→ The most important line is $r = 1$

i.e $e^{j\omega}$ which is circle (∵ radius of 1)

i.e $|z| = 1$ → unit circle (or) unit ring

Scanned with OKEN Scanner

whatever we have said in LT with reference to the jω-axis, Now we shall repeat here with reference to unit circle, inside the unit circle will be one region & outside & on circle.

→ unit circle (or ring will play a significant role in whole discussion

for given x(n), X(z) converges then

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} [x(n) r^{-n}] e^{-j\omega n}$$

$$X(z) = F[x(n) r^{-n}]$$

→ Here FT of x(n) may (or) may not exist, but FT of $x(n) r^{-n}$ may exist with proper choice of 'r'

→ so, the ROC will depend only on 'r', not on 'ω'

→ so ROC, in z-plane must be circle, because it depends only on 'r'. If 'r' is constant, if you vary 'ω' we get circle.

→ $X(e^{j\omega}) = X(z)\big|_{z=e^{j\omega}}$

** → The existance of FT can be investigated in other words as, if ROC of X(z) is given, if ROC of X(z) doesn't include the unit circle, then obviously the FT not exist.

→ ***

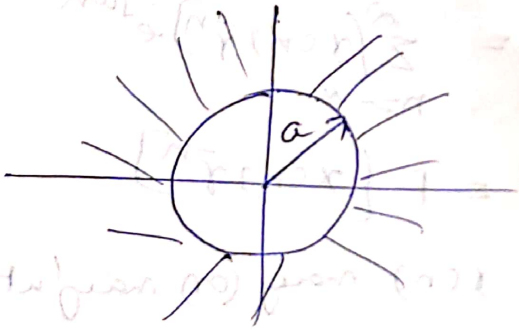
For DTFT of $x(n)$ to exist, $X(z)$ ROC must include the unit circle (i.e $|z|=1$)

Ex: $x(n) = a^n u(n)$ $[n \geq 0]$ (RSS)

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n$$ will be convergent provided $|az^{-1}|<1$ i.e $|z|>a$

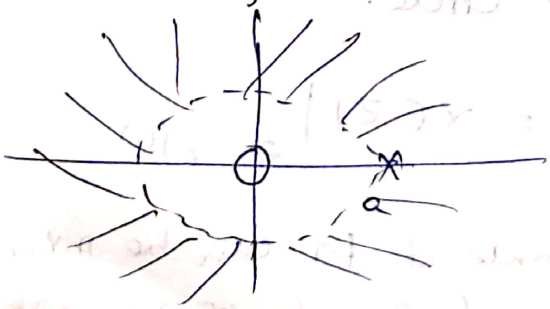
$$= \frac{1}{1-az^{-1}} \; ; \; |z| > |a|$$ ↳ radius



$$X(z) = \frac{1}{1-az^{-1}}$$ rational fn in z^{-1}

$$= \frac{z}{z-a}$$ 'rational function in z'

→ Any rational fn, can be characterized by the zero plot



Conclusion:

→ For RSS, ROC is which extend to '∞'

or $b < |z| \leq \infty$

→ ROC cannot include Poles

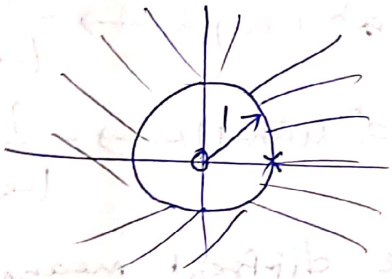
$$x(n) = a^n u(n) \longleftrightarrow \frac{1}{1-az^{-1}} \quad |z| > |a|$$

Special

$a = 1$

$$x(n) = u(n) \longleftrightarrow \frac{1}{1-z^{-1}} \quad |z| > 1$$

Roc of $u(n)$ is outside of unit circle



Ex 4 $x(n) = -a^n u(-n-1)$;

$$X(z) = \sum_{n=-\infty}^{\infty} -a^n z^{-n} = \sum_{n=-\infty}^{-1} -a^n z^{-n}$$

Let $n = n'$

$$= \sum_{-n'=-1}^{-n'=-\infty} -a^{-n'} z^{n'}$$

$$= -\sum_{n=1}^{\infty} (a^{-1}z)^n$$

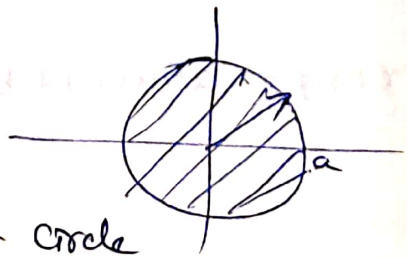
$$X(z) = -\sum_{n=1}^{\infty} (a^{-1}z)^n$$

Let's include $n=0$ also

$$= 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n$$

$$= 1 - \frac{1}{1-a^{-1}z} \quad |a^{-1}z| < 1$$

$$|z| < |a|$$



→ For LSI, Roc is inside a circle

$$X(z) = 1 - \frac{a}{a-z}$$

$$= \frac{z}{z-a} \quad ; |z| < |a|$$

$$X(z) = \frac{(a^n)}{1-az^{-1}} \quad ; |z| < |a|$$

$$a^n u(n) \longleftrightarrow \frac{1}{1-az^{-1}} \quad ; |z| > |a|$$

$$-a^n u(-n-1) \longleftrightarrow \frac{1}{1-az^{-1}} \quad ; |z| < |a|$$

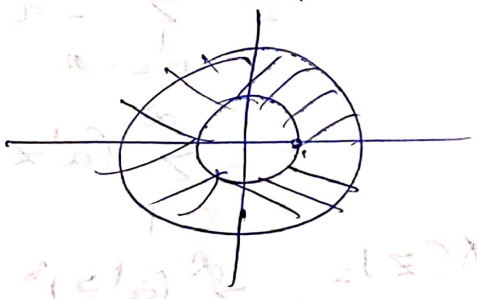
Conclusion

→ RoC are different means, sequences is different

→ RHS : RoC is $|z| > r_1$

→ LHS : RoC is $|z| < r_2$

→ If sequence is two sided ; in btw two, provided $r_2 > r_1$ { if $r_1 > r_2$, the z not exist }



$$r_1 < |z| < r_2$$

Z-Transform is linear operation

ie

$$x(n) = a x_1(n) + b x_2(n)$$

$$X(z) = a X_1(z) + b X_2(z)$$

$$\text{RoC} = \text{atleast} \{R_1 \cap R_2\}$$

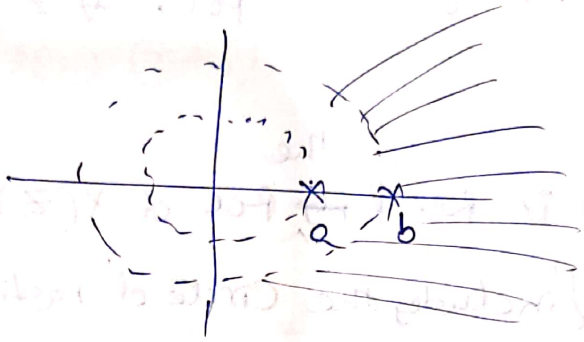
Ex: $x(n) = a^n u(n) + b^n u(n)$; (a & b are real & +ve & b > a)

$$X(z) = \frac{1}{1-az^{-1}} + \frac{1}{1-bz^{-1}}$$

ROC: $|z| > |a|$ $|z| > |b|$

$\downarrow$

$|z| > |b|$



Properties of ROC

(1) ROC : ring.

(2) ROC : Can't include a pole

(3) If it is finite duration sequence

→ for finite duration of sequence, always $z \neq$ Exist

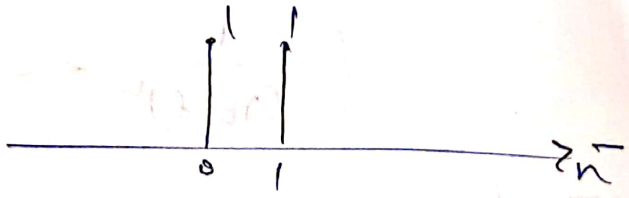
There is no Question of NonExist

ROC : $\forall z$ Except $z=0$ and / or $z=\infty$

(i.e Either one (or) both must be Excluded)

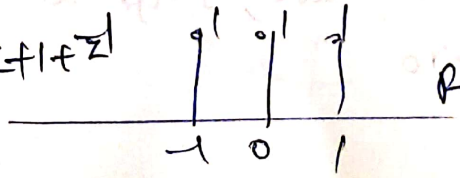
Ex: Suppose we have a RoS

Case (i)



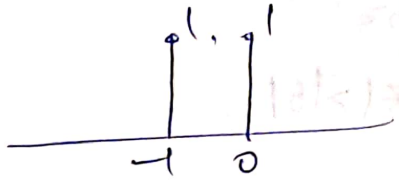
Case (ii) $X(z) = 1 + z^{-1}$ ROC: $\forall z$, Except $z=0$

$X(z) = z + 1 + z^{-1}$



ROC: $\forall z$, Except $z=0$ & $z=\infty$

Case (iii) RSS



$X(z) = z+1$

ROC; $\forall z$; Except $z=\infty$

(4)

If $x(n)$ is RSS then the ROC of $X(z) = Z[x(n)]$ includes $|z| = r_0$ [includes the circle of radius r_0] then the ROC must include all z such that $|z| > r_0$

(or)

If $x(n)$ is RSS, then ROC is outside of unit circle

Proof:

Given;

$X(z)$ converges

$|z| = r_0$

or

$$\sum |x(n) r_0^{-n}| < \infty$$

[since $x(n)$ is RSS; lets take r_0 to ∞

r_0 is >0 (or) <0]

Suppose case is $r_0 \geq 0$, then

(since RSS may start at true value (or) for -ve value)

$$\sum_{n=n_0}^{\infty} |x(n)(r_0+\delta)^{-n}| \quad \text{---(1)}$$

$\delta > 0$

$\left(\frac{1}{(r_0+\delta)^n} < \frac{1}{r_0^n} \right)$ so (1) is $< \infty$

→ If $r_0 \geq 0$ [non negative], then for all 'δ',

$\sum x(n)(r_0+\delta)$ is summable, says that it includes all values of $r > r_0$

Scanned with OKEN Scanner

any problem, if r_0 is -ve (i.e. $r_0 < 0$)

If $r_0 = -m$

$$\sum_{n=-m}^{\infty} |x(n)(r_0+\delta)^{-n}|$$

$$= \sum_{n=-m}^{-1} + \sum_{n=0}^{\infty}$$

$\sum_{n=0}^{\infty}$ → already shown $< \infty$

$\sum_{n=-m}^{-1}$ → what about this, it is finite quantity so $< \infty$

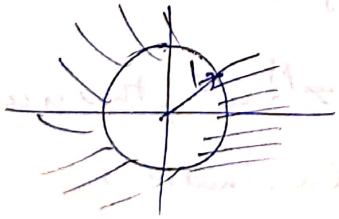
$= < \infty + < \infty$

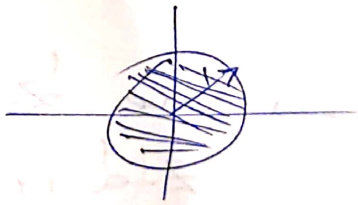
$= < \infty$

(5) If $x(n)$ is L.S. then ROC of $X(z)$ includes $|z| = r_0$ then the ROC extends to all z such that $|z| < r_0$

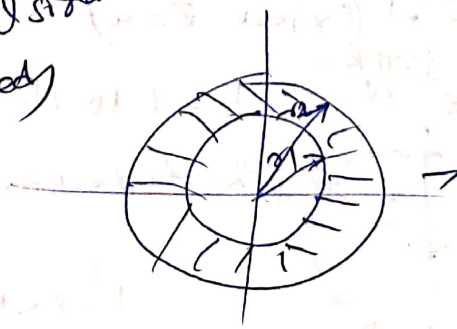
Ex: $x(n) = u(n)$; $x(n) = -u(-n-1)$

$X(z) = \frac{1}{1-z^{-1}}$; $|z| > r_0$; $X(z) = \frac{1}{1-z^{-1}}$; $|z| < 1$ ↳ r_0





(6) → ROC for bi-directional signal is finite (bounded)



→ 'r_1' is the boundary of the left

'r_2' is the boundary of Right

(6) Finite duration sequence

$x(n) = a^n$; $0 \le n \le N-1$ for $a > 0$

$= 0$; otherwise

ROC of this, $\forall z$; except '0'

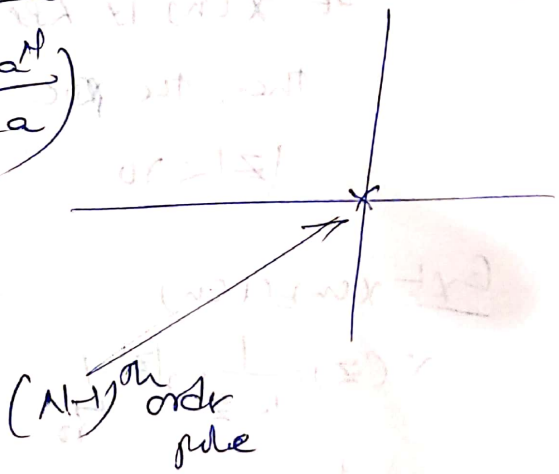
$$X(z) = 1 + az^{-1} + \cdots + a^{N-1} z^{-(N-1)}$$

$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n}$$

$$= \frac{1 - (az^{-1})^N}{1 - az^{-1}}$$

$$= \frac{z^N}{z^N} \cdot \frac{1 - a^N z^{-N}}{1 - az^{-1}}$$

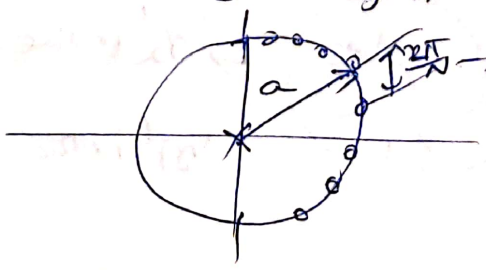
$$= \frac{1}{z^{N-1}} \left(\frac{z^N - a^N}{z - a} \right)$$



→ Since $z-a$ is a factor in $z^N - a^N$, this is why $z-a$ is not a pole (since cancelling)

$z^N = a^N \cdot 1$ (except $z = a$)

$z = a e^{\frac{j2\pi k}{N}}$; $k = 1$ to $N-1$; not $k = 0$

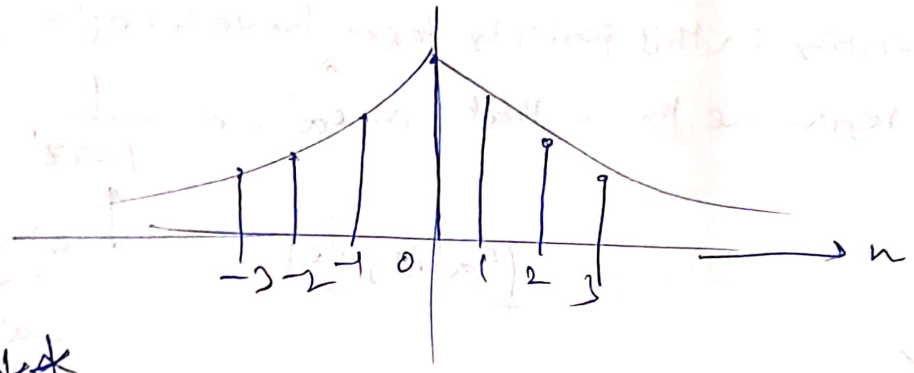


Angle difference b/w two zeros

1 is not a pole but at $e^{\frac{j2\pi k}{N}}$

Ex: Two sided signal

$x(n) = b^{|n|}$; $b>0$ & $0<b<1$



→ If $b>1$, the Z.T, blow out

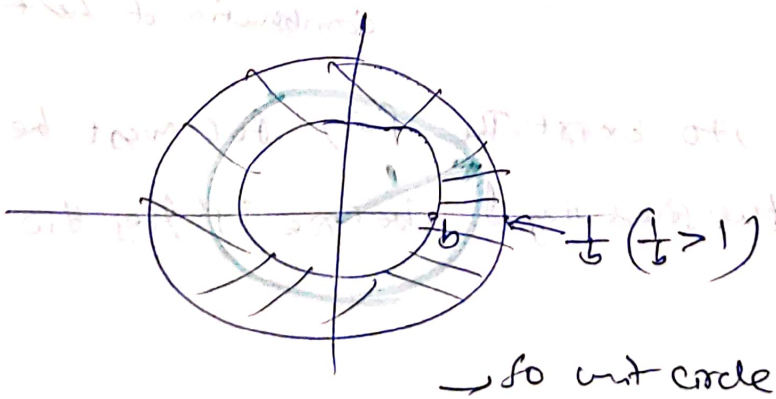
$x(n) = b^{|n|} = b^n u(n) + b^{-n} u(-n-1)$

(Rs) (Ls)

$X(z) = \frac{1}{1-bz^{-1}} + \frac{1}{1-b^{-1}z^{-1}}$

$|z| > |b|$ → $|z| < |b^{-1}|$ (or) $|z| < \frac{1}{|b|}$

$b<1$; so $\frac{1}{b} > 1$



→ so unit circle is inside the region of convergence. so $b^{|n|}$ FT exist

→ If $b>1$; Z.T not exist, there, no common region exist

$$X(z) = \frac{1}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-2z^{-1}\right)}$$

in polynomial in z^{-1}

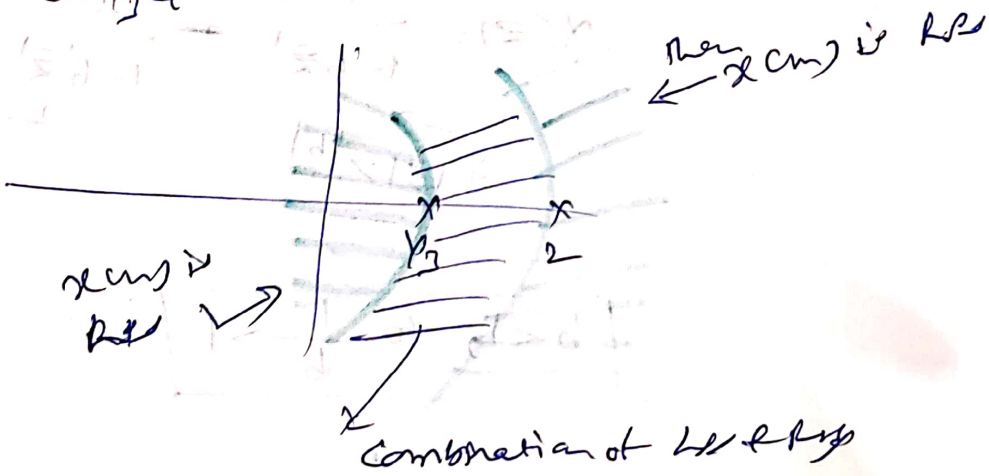
writing in this particular form we have a region, the region we know that expression of $\frac{1}{1-az^{-1}}$ ↕ anticausal (or) anti-causal

'a' is pole

→ If $X(z)$ is given in ratio of polynomials in z^{-1}, then we have to convert into z^{-1}

Possible ROC's

$a = \frac{1}{2}$ & $a = 2$



→ Here, to exist the FT, ROC must be (or) signal is two sided signal whose ROC lies between $\frac{1}{2}$ & 2

Inversion

Given $X(z)$, $x(n) = ?$

$$X(z) = X(re^{j\omega})$$

(If ω are continuous variable but r are discrete)

$$= \sum x(n) r^n e^{-jn\omega}$$

$$= F[x(n) r^n]$$

$$x(n) \longleftrightarrow X(\omega)$$

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{jn\omega} d\omega$$

$$x(n) r^n = F^{-1}[X(re^{j\omega})]$$

$$= \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) e^{jn\omega} d\omega$$

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) r^n e^{jn\omega} d\omega$$

$$= \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) (re^{j\omega})^n$$

$$= \frac{1}{2\pi} \int_{2\pi} X(z) z^n$$

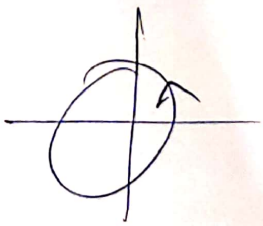
$$z = re^{j\omega}$$

$$dz = jr e^{j\omega} d\omega$$

$$d\omega = \frac{dz}{jz}$$

$$= \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$\Gamma : |z| = r$



Scanned with OKEN Scanner

$$x(n) = \frac{1}{2\pi j} \oint_{\Gamma} X(z) z^{n-1} dz$$

$\Gamma : |z| = |r|$ by is within circle

→ If $X(z)$ is rational, then we not go for process but in general if $X(z)$ is rational, then expand in partial fraction

Process: (If $X(z)$ is given in rational, in terms of z, then this trick)

$$\frac{X(z)}{z} = \sum \frac{A_i}{z - z_i}$$

$$X(z) = \sum \frac{A_i z}{z - z_i}$$

$$= \sum \frac{A_i}{1 - z_i z^{-1}}$$

$$=$$

By

$$X(z) = \frac{1 - \frac{1}{2} z^{-1}}{1 + \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}} ; \quad |z| > \frac{1}{2}$$

$$= \frac{1 - \frac{1}{2} z^{-1}}{\left(1 + \frac{1}{2} z^{-1}\right)\left(1 + \frac{1}{4} z^{-1}\right)}$$

(By observation)

$$= \frac{A}{\left(1 + \frac{1}{2} z^{-1}\right)} + \frac{B}{\left(1 + \frac{1}{4} z^{-1}\right)}$$

$$= \frac{4}{1 + \frac{1}{2} z^{-1}} - \frac{3}{1 + \frac{1}{4} z^{-1}} ; \quad |z| > \frac{1}{2}$$

Scanned with OKEN Scanner

$x(n) = 4\left(\frac{1}{2}\right)^n u(n) + 3\left(\frac{1}{4}\right)^n u(n)$

If ROC : $\frac{1}{4} < |z| < \frac{1}{2}$

$x(n) = -4\left(\frac{1}{2}\right)^n u(-n-1) - 3\left(\frac{1}{4}\right)^n u(n)$

Trick

$$X(z) = \frac{P(z)}{Q(z)} = \frac{P_1(z^{-1})}{Q_1(z^{-1})}$$

Express in the form

$\cdots + a_1 z + a_0 + a_1 z^{-1} + \cdots$, then obviously

$x(n)$ is known

Ex:

① $X(z) = \frac{1}{1-az^{-1}}$; $|z| > |a|$ → R.O.C.

$= a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots$

$a(1-az^{-1})^{-1}$

$x(0)\ x(1)\ x(2)$

$1 + az^{-1} + a^2 z^{-2} + \cdots$

$(1-az^{-1}) \overline{)\ 1}$

$1 - az^{-1}$

$x(n) = a^n u(n)$

② $\frac{1}{1-az^{-1}}$; $|z| < |a|$

$= a_0 + a_1 z + a_2 z^2 + \cdots$

To get the z^{-1}, we need to shift

$-az^{-1}+1 \overline{)\ -az^{-1} - a^2 z^{-2} - \cdots}$

1

$x(n) = -a^n u(-n-1)$

properties of Z-T:

- Linearity $x_1(n) \xrightarrow{R_1} X_1(z)$, $x_2(n) \xrightarrow{R_2} X_2(z)$

$$a x_1 + b x_2 \longleftrightarrow a X_1 + b X_2$$

R = at least $R_1 \cap R_2$

Ex:

$a^n u(n)$ $\longleftrightarrow$ $\frac{1}{1-az^{-1}}$, $|z| > |a|$

$a^n u(n-1)$ $\longleftrightarrow$ $\frac{1}{1-az^{-1}} - 1$, $|z| > |a|$

only one single x axis is to (same -1)

$a^n u(n) - a^n u(n-1) = \delta(n)$

$\frac{1}{1-az^{-1}} - \frac{1}{1-az^{-1}} - 1 = 1$

$|z| > |a|$; $|z| > |a|$ $= \forall z$

Time shifting:-

$x(n) \xrightarrow{R} X(z)$

$x(n-n_0) \longleftrightarrow z^{-n_0} X(z)$ also 'R' except for the point at origin which may have qualified

$\sum x(n-n_0) z^{-n}$

$\sum x(r) z^{-r} z^{-n_0}$

$z^{-n_0} X(z)$

$x(n-n_0) \longleftrightarrow z^{-n_0} X(z)$ → ROC = R

ROC: $R (n_0 = 0)$

If $n_0 > 0$

$\frac{X(z)}{z^{n_0}}$ If $n_0 > 0$, obviously you are creating poles at $z = 0$

∴ ROC shall be 'R' if 'R' includes the origin then we have to exclude the origin from modified region of convergence

If $n_0 < 0$; $x(n-n_0) \longleftrightarrow X(z) z^{-n_0}$

→ we are creating zeros at origin ; If there is zeros at origin there will be poles at '∞'

If $n = -2$

$X(z) z^2$ we are creating poles at '0', there may be zeros at '0' in $X(z)$, then they will cancel each other ∴ so one must be careful in specifying ROC.

Frequency shifting

$x(n) \longleftrightarrow X(z)$ → R

$e^{j\omega_0 n} x(n) \longleftrightarrow X(ze^{j\omega_0})$ → R

modulation

rotation

$\sum x(n) e^{j\omega_0 n} z^{-n}$

$= \sum x(n) (e^{j\omega_0} z)^{-n}$

If $z = re^{j\omega}$

$X(re^{j(\omega - \omega_0)})$

∴ the angle gets rotated in the clockwise direction by ω_0. So $X(ze^{j\omega_0})$ stands for rotation

Scanned with OKEN Scanner

(10)

→ ROC never change, because ROC depends on the distance from the origin (i.e r) not an angle

In general

$x(n) \longleftrightarrow X(z)$ → R

$z_0^n x(n) \longleftrightarrow X\left(\frac{z}{z_0}\right)$

(z_0 is constant)

z_0 is complex quantity

z_0 will be in the format

$\boxed{z_0 = r_0 e^{j\omega_0}}$

RZ_0

$X(z)$ has a pole at $z=p$

then

$X\left(\frac{z}{z_0}\right)$ " " " $z = pz_0 = (z_0)p$

scaled by z_0

Time reversal:-

$x(n) \longleftrightarrow X(z)$ → R

$x(-n) \longleftrightarrow X\left(\frac{1}{z}\right)$ → $\frac{1}{R}$

$\sum x(-n) z^{-n} = \sum x(n) \left(\frac{1}{z}\right)^{-n}$

$= X\left(\frac{1}{z}\right)$

i.e. → If z_0 (point) is in 'R'

Then $\frac{1}{z_0}$ is in ROC of $\frac{1}{R}$

Scanned with OKEN Scanner

Convolution

$$Z\{x(n) * h(n)\} = X(z)\,H(z)$$

R_1, R_2

R = atleast $R_1 \cap R_2$

Differentiation in the z-domain

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$-z\frac{d}{dz}X(z) = \sum_{n=-\infty}^{\infty} (+n)\,x(n)\,z^{-n-1}$$

$$-z\frac{d}{dz}X(z) \longleftrightarrow n\,x(n)$$

atleast 'R' (it may be wider)

Ex1 $X(z) = \log z$

$-z\frac{1}{z} = -1$ all z

application of above

$$x(n) \longleftrightarrow X(z) = \log(1+az^{-1})$$

to find z-T, it must be analog

$|z| > |a|$

$$-z\frac{dY}{dz} = \frac{az^{-1}}{1+az^{-1}} \longleftrightarrow n\,x(n)$$

$$u[n] \longleftrightarrow \frac{1}{1-az^{-1}}$$

$$\frac{1}{1-az^{-1}} \longleftrightarrow a^n u(n)$$

$$\frac{1}{1+az^{-1}} \longleftrightarrow (-a)^n u(n)$$

$$\frac{a}{1+az^{-1}} \longleftrightarrow a(-a)^n u(n)$$

delayed by 1

$$\frac{az^{-1}}{1+az^{-1}} \longleftrightarrow a(-a)^{n-1}u(n-1) = n\,x(n)$$

Scanned with OKEN Scanner

(4) $x(n) = \frac{a(-a)^{n-1}u(n-1)}{n}$

$= -\frac{-a(-a)^{n-1}u(n-1)}{n}$

$x(n) = -\frac{(-a)^n u(n-1)}{n}$

(2) $\frac{az^{-1}}{(1-az^{-1})^2}$ (2) → due to differentiation

what $X(z) = \frac{1}{1-az^{-1}} \longleftrightarrow a^n u(n)$

$-z\frac{d}{dz}\left(\frac{1}{1-az^{-1}}\right) = \frac{az^{-1}}{(1-az^{-1})^2} \longleftrightarrow na^n u(n)$

initial value theorem:- (IVT)

→ applies for a sequence for $x(n) = 0;\ n<0$

Then I.V $x(0)$.

If $x(n) = 0,\ n<0$, then,

$X(z) = x(0) + x(1)z^{-1} + x(2)z^{-2}$

$+\infty \quad +\infty$

$x(0) = \lim_{z\to\infty} X(z)$ provided the RoC

→ suppose it was not RoC, if it is Less than initial value theorem, cannot have been defined

Then $x(0) = \lim_{z\to 0} X(z)$

$$x(0) = \lim_{z \to \infty} X(z)$$

→ If $X(z) = \frac{P(z)}{Q(z)}$ = rational fn

If $x(0)$ is finite then $P^0 \leq Q^0$

Interpolation (samples are created)

$$w(n) = \begin{cases} x(r), & n = kr \\ 0, & \text{otherwise} \end{cases}$$

↑ derived from $x(n)$

If $k=3$ then

$w(0) = x(0)$

$w(1) = 0$; (∵ '1' is not multiple of 3)

$w(2) = 0$;

$w(3) = x(1)$

Decimation :- (by dividing by a factor of '10',)

but in general in Electrical engg dividing by any factor gets a name called

Let $k=3$

$$W(z) = w(0) + w(1)z^{-1} + w(2)z^{-2} + w(3)z^{-3} + \cdots$$

$$= x(0) + x(1)z^{-3} + x(2)z^{-6} + \cdots$$

$$= X(z^3)$$

If k is general quantity then

$$w(z) = X(z^k)$$

Scanned with OKEN Scanner